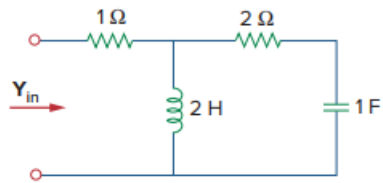


Problem

At $\omega = 1$ rad/s, obtain the input admittance in the circuit of Fig. 9.64.

Figure 9.64:



Step-by-step solution

Step 1 of 3

Refer to figure 9.64 given in the textbook.

Redraw the circuit for $\omega = 1$ rad/s as shown in figure 1.

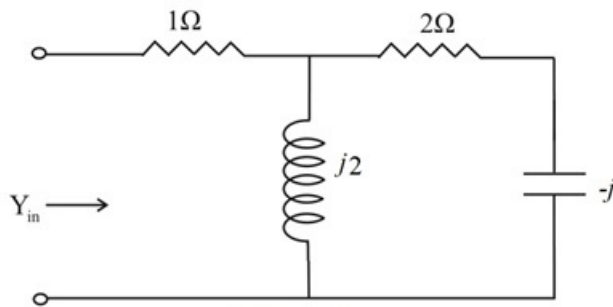


Figure 1

Step 2 of 3

Calculate the impedance Z_{in} seen from the same end from which the admittance is seen.

$$Z_{in} = 1 + [j2 \parallel (2 - j)]$$

$$Z_{in} = 1 + \left[\frac{j2(2 - j)}{j2 + 2 - j} \right]$$

$$Z_{in} = 1 + \left[\frac{2 + j4}{2 + j} \right]$$

$$Z_{in} = 1 + (1.6 + j1.2)$$

$$Z_{in} = 2.6 + j1.2 \, \Omega$$

Step 3 of 3

Calculate the admittance Y_{in} .

$$Y_{in} = \frac{1}{Z_{in}}$$

Substitute $2.6 + j1.2$ for Z_{in} .

$$Y_{in} = \frac{1}{2.6 + j1.2}$$

Therefore, the admittance Y is $\boxed{0.3171 - j0.1463 \text{ S}}$.